University of Illinois
ECE 310 **Profs. Liang and Snyder**

Midterm Exam 1

7:00-9:00pm, Wednesday, February 26, 2025

Name: Solutions

NetID: _____

Score: _____

Problem	Pts.	Score
1	12	
2	12	
3	10	
4	16	
5	18	
6	6	
7	6	
8	20	
Total	100	

Instructions

- You may not use any books, calculators, or notes other than one handwritten two-sided sheet of 8.5" x 11" paper.
- Show all your work to receive full credit for your answers.
- When you are asked to "calculate", "determine", or "find", this means providing closed-form expressions (i.e., without summation or integration signs).
- Neatness counts. If we are unable to read your work, we cannot grade it.
- Turn in your entire booklet once you are finished. No extra booklet or papers will be considered.

(12 Pts.)

1. Mark "**True**" or "**False**" for the following statements.
   Grading: Correct answer = 2 pt.; Incorrect answer = -1 pt. No answer = 0 pts.

   (a) If the output, $y[n]$, of a system is related to its input, $x[n]$, by $y[n] = \sum_{\ell=-\infty}^{\infty} h[\ell]x[n-\ell]$, for some well-defined function $h[n]$, the system must be a causal system. **T/F**

   (b) If the unit pulse response, $h[n]$, of a system $\mathcal{S}$ is absolutely summable, i.e., $\sum_{n=-\infty}^{\infty} |h[n]| < \infty$, then $\mathcal{S}$ must be BIBO-stable regardless if $\mathcal{S}$ is LSI or not. **T/F**

   (c) The z-transform of $x[n] = e^{j\frac{\pi}{4}n}$ does not exist because the ROC is empty. **T/F**

   (d) The parallel connection of two BIBO stable LTI systems always results in another BIBO stable LTI system. **T/F**

   (e) A BIBO stable LTI system has a transfer function given by $H(z) = \frac{1}{1-\sqrt{2}z^{-1}}$. This system must be anti-causal. **T/F**

   (f) An LTI system with a two-sided impulse response is never BIBO stable. **T/F**

(d) $X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$

$= \sum_{n=-\infty}^{\infty} e^{j\frac{\pi}{4}n} z^{-n}$

$= \left(\sum_{n=-\infty}^{-1} \left(\frac{e^{j\frac{\pi}{4}}}{z}\right)^n\right) + \left(\sum_{n=0}^{\infty} \left(\frac{e^{j\frac{\pi}{4}}}{z}\right)^n\right)$

$n = -m-1$

$= \left(\sum_{m=0}^{\infty} \left(\frac{e^{j\frac{\pi}{4}}}{z}\right)^{-m-1}\right) + \frac{1}{1-e^{j\frac{\pi}{4}}z^{-1}}$

if $|z|>1$, else sum does not converge.

$= \left(\frac{e^{j\frac{\pi}{4}}}{z}\right)^{-1} \sum_{m=0}^{\infty} \left(\frac{z}{e^{j\frac{\pi}{4}}}\right)^m$

$= e^{-j\frac{\pi}{4}} z^{-1} \left(\frac{1}{1-e^{-j\frac{\pi}{4}}z}\right)$

if $|z|<1$, else sum diverges

Overall ROC is the $|z|>1 \cap |z|<1$, which is empty!

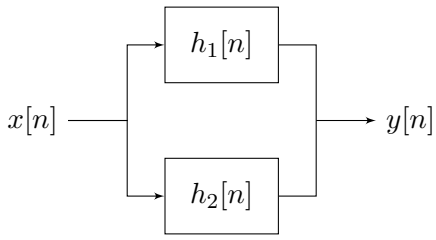
(12 Pts.)

2. For each of the systems with input $x[n]$ and output $y[n]$ shown in the table, indicate by "**yes**" or "**no**" whether the properties indicated apply to the system. **Note**: you do not need to provide proofs/justification. Points will not be deducted for incorrect answers.

	Linear	Shift-Invariant	Causal	Stable
$y[n] = x[n] * j^n u[n]$	Yes	Yes	Yes	No
$y[n] = 2x[\lvert n \rvert] + 10$	No	No	No	Yes
$y[n] = e^{x[n]+1}$	No	Yes	Yes	Yes

(10 Pts.)

3. Consider the following parallel connection of two LTI systems with impulse responses $h_1[n]$ and $h_2[n]$, respectively.



We are given that $h_1[n] = \delta[n-1]$. Suppose we know the following input-output pair $x[n]$ and $y[n]$.

$$x[n] = \{\underset{\uparrow}{1}, 2, 1\}$$
$$y[n] = \{1, \underset{\uparrow}{2}, 2, 2, 1\}$$

(a) Determine the impulse response $h_2[n]$.

or in the z-domain

$* \; y[n] = x[n] * h[n]$

$= x[n] * h_1[n] + x[n] * h_2[n]$

$= \{1, \underset{\uparrow}{2}, 2, 2, 1\}$

$* \; x[n] * h_1[n] = \{\underset{\uparrow}{0}, 1, 2, 1\}$

$* \; x[n] * h_2[n] = y[n] - x[n] * h_1[n]$

$= \{1, \underset{\uparrow}{2}, 1, 0, 0\}$

$= x[n+1]$

$\hookrightarrow \boxed{h_2[n] = \delta[n+1]}$

$Y(z) = X(z)H_1(z) + X(z)H_2(z)$

$z + 2 + 2z^{-1} + 2z^{-2} + z^{-3} = z^{-1} + 2z^{-2} + z^{-3} + (1 + 2z^{-1} + z^{-2})H_2(z)$

$z + 2 + z^{-1} = (1 + 2z^{-1} + z^{-2})H_2(z)$

$H_2(z) = z$

$\hookrightarrow h_2[n] = \delta[n+1]$

(b) Is the overall system of $h[n]$, i.e. such that $x[n] * h[n] = y[n]$, causal? Justify your reasoning.

$h[n] = h_1[n] + h_2[n] = \{1, \underset{\uparrow}{0}, 1\}$

The overall system is not causal because $h[n] \neq 0$ for all $n < 0$.

(16 Pts.)

4. Compute the response of an LSI system with impulse response $h[n]$ for a given input $x[n]$ in each of the following cases.

(a) $x[n] = \{1, -2, \underset{\uparrow}{0}, 3, -1, 1\}$, $h[n] = \{\underset{\uparrow}{2}, -1, 3\}$

* $x[n]$ starts at $n=-2$
* $h[n]$ starts at $n=0$
∴ $y[n]$ starts at $n = -2+0 = -2$.

$$y = Hx = \begin{bmatrix} 2 & 0 & 0 & 0 & 0 & 0 \\ -1 & 2 & 0 & 0 & 0 & 0 \\ 3 & -1 & 2 & 0 & 0 & 0 \\ 0 & 3 & -1 & 2 & 0 & 0 \\ 0 & 0 & 3 & -1 & 2 & 0 \\ 0 & 0 & 0 & 3 & -1 & 2 \\ 0 & 0 & 0 & 0 & 3 & -1 \\ 0 & 0 & 0 & 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \\ 0 \\ 3 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ -5 \\ 5 \\ 0 \\ -5 \\ 12 \\ -4 \\ 3 \end{bmatrix} \leftarrow n=0$$

$$y[n] = \{2, -5, \underset{\uparrow}{6}, 0, -5, 10, -4, 3\}$$

(b) $x[n] = \left(-\frac{1}{3}\right)^n u[n]$, $h[n] = \left(\frac{1}{2}\right)^n u[n-3]$

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k]$$
$$= \sum_{k=-\infty}^{\infty} \left(\frac{1}{2}\right)^k u[k-3] \left(-\frac{1}{3}\right)^{n-k} u[n-k]$$
$$= \left(-\frac{1}{3}\right)^n \sum_{k=3}^{n} \left(-\frac{3}{2}\right)^k$$

*Let $m = k-3$, $k = m+3$ →
$$= \left(-\frac{1}{3}\right)^n \sum_{m=0}^{n-3} \left(-\frac{3}{2}\right)^{m+3}$$
$$= \left(-\frac{3}{2}\right)^3 \left(-\frac{1}{3}\right)^n \sum_{m=0}^{n-3} \left(-\frac{3}{2}\right)^m$$ (this summation is empty for $n<3$)
$$= \left(-\frac{3}{2}\right)^3 \left(-\frac{1}{3}\right)^n \left(\frac{1-\left(-\frac{3}{2}\right)^{n-2}}{1-\left(-\frac{3}{2}\right)}\right) u[n-3]$$
$$= -\frac{27}{8}\left(-\frac{1}{3}\right)^n \left(\frac{1-\left(-\frac{3}{2}\right)^{n-2}}{\frac{5}{2}}\right) u[n-3]$$

$$y[n] = -\frac{27}{20}\left(-\frac{1}{3}\right)^n u[n-3] + \frac{27}{20}\left(-\frac{1}{3}\right)^n \left(-\frac{3}{2}\right)^{n-2} u[n-3]$$
$$y[n] = -\frac{27}{20}\left(-\frac{1}{3}\right)^n u[n-3] + \frac{3}{5}\left(\frac{1}{2}\right)^n u[n-3]$$

$$Y(z) = X(z)H(z)$$
$$= \left(\frac{1}{1+\frac{1}{3}z^{-1}}\right)\left(\left(\frac{1}{2}\right)^3 \cdot \frac{z^{-3}}{1-\frac{1}{2}z^{-1}}\right)$$
$$= \left(\frac{1}{8}z^{-3}\right)\left(\frac{1}{(1+\frac{1}{3}z^{-1})(1-\frac{1}{2}z^{-1})}\right)$$
↓ PFD
$$= \left(\frac{1}{8}z^{-3}\right)\left(\frac{\frac{2}{5}}{1+\frac{1}{3}z^{-1}} + \frac{\frac{3}{5}}{1-\frac{1}{2}z^{-1}}\right)$$

$$y[n] = \frac{1}{20}\left(-\frac{1}{3}\right)^{n-3} u[n-3] + \frac{3}{40}\left(\frac{1}{2}\right)^{n-3} u[n-3]$$

Equivalent Answers!

(18 Pts.)

5. Compute the z-transform of each of the following signals including its ROC.

(a) $x[n] = \sum_{k=0}^{2} \left(\frac{1}{2}\right)^k u[n-k]$

Alternate valid forms of $X(z)$

* $X(z) = 1 + \frac{3}{2}z^{-1} + \frac{\frac{7}{4}z^{-2}}{1-z^{-1}}$

$$X[n] = u[n] + \frac{1}{2}u[n-1] + \frac{1}{4}u[n-2]$$

$$\boxed{X(z) = \frac{1 + \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2}}{1 - z^{-1}}, \quad |z| > 1}$$

(b) $x[n] = 3^n u[-n+2]$

* $X[n] = 3^n[-(n-2)] = 3^3 \cdot 3^{n-3} u[-(n-2)] = 27g[n-3]$
* $g[n] = 3^n u[-(n+1)]$ * $G(z) = \frac{-1}{1-3z^{-1}}, \quad |z| < 3$
* $X(z) = 27z^{-3}(-1/z)$

Alternate valid forms of $X(z)$

* $X(z) = \frac{9z^{-2}}{1-\frac{1}{3}z}$
* $X(z) = \frac{-1}{1-3z^{-1}} + 1 + 3z^{-1} + 9z^{-2}$

$$\boxed{X(z) = \frac{-27z^{-3}}{1-3z^{-1}}, \quad 0 < |z| < 3}$$

(c) $x[n] = \left(\frac{1}{4}\right)^n \left(\frac{2}{3}\right)^{n-2} u[n-1] + \left(\frac{4}{3}\right)^n u[-n-1]$

$$X[n] = \left(\frac{2}{3}\right)^{-2}\left(\frac{2}{12}\right)^n u[n-1] + \left(\frac{4}{3}\right)^n u[-n-1]$$
$$= \frac{9}{4}\left(\frac{1}{6}\right)\left(\frac{1}{6}\right)^{n-1} u[n-1] + \left(\frac{4}{3}\right)^n u[-n-1]$$

$$X(z) = \underbrace{\frac{\frac{3}{8}z^{-1}}{1-\frac{1}{6}z^{-1}}}_{\text{ROC: } |z| > \frac{1}{6}} - \underbrace{\frac{1}{1-\frac{4}{3}z^{-1}}}_{\text{ROC: } |z| < \frac{4}{3}}$$

$$\boxed{X(z) = \frac{\frac{3}{8}z^{-1}}{1-\frac{1}{6}z^{-1}} - \frac{1}{1-\frac{4}{3}z^{-1}}, \quad \frac{1}{6} < |z| < \frac{4}{3}}$$

(6 Pts.)

6. The transfer function of a causal LSI system is given below:

$$H(z) = \frac{z}{z - \frac{1}{\sqrt{2}}(1+j)}, \quad \text{ROC: } |z| > 1$$

Mark "**True**" or "**False**" for each of the below inputs that will produce an **unbounded** output.

Pole @ $z = \frac{1}{\sqrt{2}}(1+j) = e^{j\frac{\pi}{4}}$

(a) $u[n]$ — **T / F**

(b) $e^{j\frac{\pi}{4}n}u[n]$ — **T / F**

(c) $e^{-j\frac{\pi}{4}n}u[n]$ — **T / F**

(d) $e^{-j\frac{3\pi}{4}n}u[n]$ — **T / F**

(e) $\cos(\frac{\pi}{4}n)u[n]$ — **T / F**

(f) $4^n u[n]$ — **T / F**

(6 Pts.)

7. Consider a causal LSI system given by the following difference equation

$$y[n] = -\frac{3}{2}y[n-1] + y[n-2] + x[n] - \alpha^2 x[n-2],$$

where $\alpha \in \mathbb{C}$. Mark "**True**" or "**False**" for each of the below values for α that will make this system BIBO stable.

$Y(z)\left(1 + \frac{3}{2}z^{-1} - z^{-2}\right) = X(z)\left(1 - \alpha^2 z^{-2}\right)$

$\frac{Y(z)}{X(z)} = H(z) = \frac{(1-\alpha z^{-1})(1+\alpha z^{-1})}{(1+2z^{-1})(1-\frac{1}{2}z^{-1})}$

(a) 2 — **T / F**

(b) -2 — **T / F**

(c) j2 — **T / F**

(d) -j2 — **T / F**

(e) $\frac{\sqrt{2}}{2}$ — **T / F**

(f) $\frac{1}{2}$ — **T / F**

(20 Pts.)

8. Consider an LTI system with transfer function

$$H(z) = \frac{(2 - 3z^{-1})}{\left(1 - \frac{1}{2}z^{-1}\right)\left(1 + \frac{3}{2}z^{-1}\right)}.$$

(a) Determine the difference equation corresponding to this system.

$$H(z) = \frac{Y(z)}{X(z)} = \frac{2 - 3z^{-1}}{1 + z^{-1} - \frac{3}{4}z^{-2}}$$

$$Y(z)\left(1 + z^{-1} - \frac{3}{4}z^{-2}\right) = X(z)\left(2 - 3z^{-1}\right)$$

$$\boxed{y[n] + y[n-1] - \frac{3}{4}y[n-2] = 2x[n] - 3x[n-1]}$$

(b) Determine all the possible outputs $y[n]$ when the input is $x[n] = \delta[n] + \frac{3}{2}\delta[n-1]$.

$$Y(z) = X(z)H(z) = \left(1 + \frac{3}{2}z^{-1}\right)\cdot\frac{\left(2 - 3z^{-1}\right)}{\left(1 - \frac{1}{2}z^{-1}\right)\left(1 + \frac{3}{2}z^{-1}\right)} = \frac{2 - 3z^{-1}}{1 - \frac{1}{2}z^{-1}} = \frac{2}{1 - \frac{1}{2}z^{-1}} - \frac{3z^{-1}}{1 - \frac{1}{2}z^{-1}}$$

(#1) $y[n] = 2\left(\frac{1}{2}\right)^n u[n] - 3\left(\frac{1}{2}\right)^{n-1} u[n-1]$ ← Pole is right-sided

(#2) $y[n] = -2\left(\frac{1}{2}\right)^n u[-n-1] + 3\left(\frac{1}{2}\right)^{n-1} u[-n]$ ← Pole is left-sided

(c) Determine the ROC of $H(z)$ so that the system is stable and find the corresponding the impulse response $h[n]$.

* Pole @ $z = \frac{1}{2}$ is right-sided
* Pole @ $z = -\frac{3}{2}$ is left-sided

→ $\boxed{\text{ROC} = \frac{1}{2} < |z| < \frac{3}{2}}$

$$H(z) = \frac{2 - 3z^{-1}}{\left(1 - \frac{1}{2}z^{-1}\right)\left(1 + \frac{3}{2}z^{-1}\right)} = \frac{A_1}{1 - \frac{1}{2}z^{-1}} + \frac{A_2}{1 + \frac{3}{2}z^{-1}}$$

$$2 - 3z^{-1} = A_1\left(1 + \frac{3}{2}z^{-1}\right) + A_2\left(1 - \frac{1}{2}z^{-1}\right)$$

A_1: $z = \frac{1}{2}$: $-4 = 4A_1$, $A_1 = -1$

A_2: $z = -\frac{3}{2}$: $4 = \frac{4}{3}A_2$, $A_2 = 3$

$$\boxed{h[n] = -\left(\frac{1}{2}\right)^n u[n] - 3\left(-\frac{3}{2}\right)^n u[-n-1]}$$

Scratch paper: Do not remove from the exam!